Brownian Motion

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What is Brownian Motion?

A vast part of probability theory is devoted to describing the macroscopic picture emerging from random systems defined by microscopic phenomena. Brownian motion is the macroscopic picture arising from a particle moving randomly in a *d*-dimensional space without making large jumps.

On the microscopic level, at any time step, the particle receives a random displacement, cased by other particles hitting it or by an external force, so that, if its position at time 0 is x_0, then its position at time n is given as $x_n = x_0 + \sum_{i=1}^{n} x_i$, where the displacements $x_1, x_2, \ldots$ are assumed to be IID random variables with values in $\mathbb{R}^d$. The process $\{x_n\}_{n\geq 0}$ is a random walk, and the displacements represent the microscopic inputs. When referring to the macroscopic picture, we want to know:

- If x_n drifts to infinity.
- If x_n returns to the neighbourhood of origin infinitely often.
- The speed of growth of $\max\{|x_1|, \ldots, |x_n|\}$ as $n \to \infty$.

Thus, leading to definition of Brownian motion. Do note that a standard Brownian motion, $B(t)$ or B_t, can be denoted by $W(t)$ since it is a Wiener process. A real-valued stochastic process $\{B(t) : t \geq 0\}$ is called a (linear) Brownian motion with start in $x \in \mathbb{R}$ if the following holds:

- $B(0) = x$;
- The process has independent increments, *i.e.*, $\forall\ 0 \leq t_1 \leq t_2 \leq \ldots \leq t_n$ the increments $B(t_n) - B(t_{n-1}), B(t_{n-1}) - B(t_{n-2}), \ldots, B(t_2) - B(t_1)$ are independent R.V.s;
- $\forall\, t \geq 0, h > 0$ the increments $B(t+h) - B(t)$ are normally distributed with expectation 0 and variance h, *i.e.*, the process has stationary increments;
- Almost surely, the function $t \to B(t)$ is continuous.

We say that $\{B(t) : t \geq 0\}$ is a standard Brownian motion if $x = 0$. In 1923, Wiener proved that standard Brownian motion exists. There are a few important properties of Brownian motion, that is: it is continuous, $\mathbb{E}[B(t)] = 0, \mathbb{E}[B(t)^2] = t, B(t) \sim N(0,t)$, and in continuous space, a Brownian motion has the martingale (that is, $\mathbb{E}[B(t_i + t_j) \mid B(t_i)] = B(t_i)$) and Markov property. If $i < j$, then $Cov(B(t_i), B(t_j)) = \mathbb{E}[B(t_i)B(t_j)] = \mathbb{E}[B(t_i)(B(t_j) - B(t_i))] + \mathbb{E}[B(t_i)^2] = t_i$, thus, $Cov(B(s), B(t)) = \min\{s,t\}$.

Since $Var(B(t)) = t$, then the correlation coefficient is $Corr(B(s), B(t)) = \frac{Cov(B(s),B(t))}{\sqrt{Var(B(s))Var(B(t))}} = \frac{\min\{s,t\}}{\sqrt{st}}$. Without loss of generality, if $s \leq t$, then $Corr(B(s), B(t)) = \sqrt{\frac{s}{t}}$. As $t \to \infty$, with s fixed, $Corr(B(s), B(t)) \to 0$, showing that there is a weakening of dependence between the process at widely separated times.

An Example on Brownian Motion

Let B_t be a Brownian motion. What is the probability that $B_1 < 0$ and $B_2 > 0$?

Solution: Since Brownian motion has independent increments, we have that $B_1 \sim N(0,1)$ and $B_2 - B_1 \sim N(0,1)$, where these random variables are independent. Note that the event $\{B_1 < 0, B_2 > 0\}$ may be rewritten as $\{B_1 < 0, B_2 - B_1 > -B_1\}$. Since $B_1 < 0$, we have $-B_1 = |B_1|$, and furthermore, if $B_2 - B_1 > -B_1 > 0$, that is $B_2 - B_1 > 0$. Therefore, $\{B_1 < 0, B_2 > 0\} = \{B_1 < 0, B_2 - B_1 > 0, |B_2 - B_1| > |B_1|\}$. Now, $P(B_1 < 0) = \frac{1}{2}, P(B_2 - B_1 > 0) = \frac{1}{2}$. Conditional on the event $\{B_1 < 0, B_2 - B_1 > 0\}$, the pair $(B_1, B_2 - B_1)$ lies in the second quadrant, and because B_1 and $B_2 - B_1$ are independent standard normal random variables, their joint density is rotationally symmetric about the origin. The line $y = -x$ bisects the second quadrant into two sectors of equal angle. Hence, within the second quadrant, $\mathbb{P}(|B_2 - B_1| > |B_1 \mid B_1 < 0, B_2 - B_1 > 0) = \frac{1}{2}$. Therefore, $P(B_1 < 0, B_2 > 0) = \left(\frac{1}{2}\right)\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = \frac{1}{8}$. The final $\frac{1}{2}$ arises from the symmetry of the joint Gaussian density and should be interpreted as a conditional probability, rather than as a consequence of independence.

Conclusion

Brownian motion is arguably the most fundamental stochastic processes in probability theory and mathematical finance. It happens naturally as the scaling limit of random walks, and provides a mathematical model for the cumulative effect of many small random fluctuations. Its properties include independent and stationary increments, continuity of paths, the Markov property, and the martingale property. The example used in this article can illustrate how these properties can be combined with geometric symmetry arguments to compute

probabilities elegantly. Brownian motion leads into more advanced topics such as Ito calculus, stochastic differential equations, option pricing, *etc.*

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